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Analysis of the best model.

The dataset was divided into three parts: training, validation, and test data.

The models were trained using the training data and evaluated using validation and test data to check how well they perform on new data.

Models Used: Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, K-Nearest Neighbors (KNN), Support Vector Classifier (SVC).

The evaluation metrics are: Accuracy, Precision, Recall, F1 Score, ROC-AUC.

From the results, some models like Random Forest, Gradient Boosting, and KNN showed very high accuracy (around 0.95). However, their precision, recall, and F1-score were close to 0. This means that even if accuracy is high, these models are not correctly identifying the positive class. This problem happens because the dataset is likely imbalanced.

The best performing models are:

Logistic Regression

Logistic Regression performed well compared to other models. It had good ROC-AUC (around 0.83) and high recall (around 0.76). It was also consistent in both validation and test datasets.

Support Vector Classifier (SVC)

SVC also performed well, especially in recall (above 0.80), meaning it can detect most of the positive cases. However, its precision is lower, so it makes more false positive predictions.

The voting / averaging

The best three models selected were:

- Logistic Regression
- Gradient Boosting
- SVC

By combining these models using voting (average probabilities), the performance became more stable. It improved the balance between recall and ROC-AUC.

Final ensemble

The ensemble model gave better overall performance compared to single models. It combines strengths of different models and reduces their weaknesses.

Final analyses

Even though some models showed very high accuracy, they are not useful because they fail to detect positive cases.

The best individual model is Logistic Regression because, it has a good balance between metrics, detects positive cases better and performs consistently.

SVC is also a strong model, especially for recall.

The ensemble model is the best overall because it combines multiple models and gives more reliable results.

Limitation:

One important issue in this project is class imbalance, this caused some models to predict mostly one class and ignore the other.

Conclusion:

In conclusion, accuracy alone is not enough to evaluate models in this case, logistic regression and SVC performed better because they can detect positive cases.

Even though Logistic Regression is the best individual model, overall the ensemble model gave the best results.